

Printed surface texture and the electrical output of a small vertical-axis wind turbine rotor

Wind-tunnel results, August–September 2026

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Data URI_VAWT_Roughness_Data_2026-09-30.zip (accompanies this report)

Summary

Bottom line. On one mounting each, rotors printed with 0.10 and 0.20 mm of fuzzy-skin texture produced about 8% and 14% more peak electrical power than the 0.05 mm rotor across 10.2–38.0 m/s. The extra power comes with the rotor turning faster in the same wind, not from the electrical side. It is not yet shown that texture, rather than mounting, print differences or a change between the first run and the others, is the cause. The remounts planned for 1 October are the first check.

Rotor (label)	Fuzzy-skin thickness	$P_{\max}$ at 38.0 m/s	Change vs Ra 20	95% CI*
Ra 20	0.050 mm	3.79 W	—	—
Ra 40	0.101 mm	4.10 W	+8.4%	[+5.7, +11.2]
Ra 80	0.202 mm	4.50 W	+13.7%	[+6.9, +21.0]

*Change is the geometric mean over 14 wind speeds. The interval covers scatter between wind speeds within a single mounting, allowing for correlation between neighbouring points. It does not cover mount-to-mount variation.

- **Consistency.** Ra 80 is higher than Ra 20 at all 14 wind speeds, and Ra 40 at 13, the fourteenth being equal to within 0.1 mW (Fig. 1). Ra 80 is higher than Ra 40 by +4.9% ([+1.5, +8.5]). Ra 40 and Ra 80 were both tested after a round of software and firmware changes, so that comparison is free of the one confound that sets Ra 20 apart (Section 7).
- **Mechanism.** The textured rotors' open-circuit voltage, which tracks free-running rotor speed, is higher at every wind speed (+4.1% and +6.4%), while the apparent source resistance is unchanged (Section 5.4). For Ra 80 the advantage grows with wind speed, from about +3.5% below 17 m/s to +7.8% above 31 m/s.
- **Caveats.** Each rotor was mounted once. The Ra labels are targets estimated when the prints were planned, never measured (Section 2). The rig measures electrical power, not the aerodynamic power coefficient, and it did not record rotor speed.

Responses to open requests.

- *Comparison with the initial test (Prof. Jeong, 27 July).* Over 800–1700 rpm the 5 June values lie between −6.6% and +25.3% of the rig's Ra 20 (geometric mean +6.2%), within the range the three textured rotors span (Section 6). The rotor and the processing differ, so this is a consistency check, not a texture comparison.
- *The 27 July no-texture test.* Reprocessed from its raw export, it cannot resolve a difference of 8–14% (Section 6). Running the same rotor on the rig is the direct comparison.

- *Which blade exactly (Prof. Jahangiri, 27 July)*. Section 2 lists what is known about each rotor, including what is still inferred.
- *Fault-testing plan (Prof. Jeong, 15 and 27 July)*. Not yet drafted. A starting point is proposed in Section 9.1.

Acknowledgement. Taegu Kang co-ran the 27 July, 20 August and 26 August tests, and shared the complete raw export of the July test, without which Section 6 would not have been possible.

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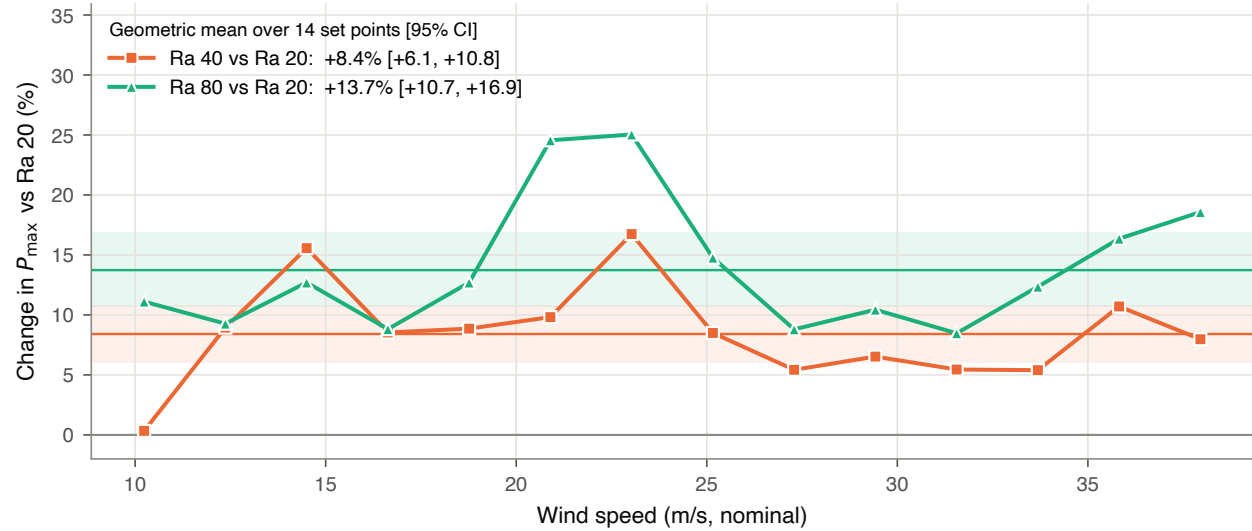


Figure 1. Change in peak electrical power (raw arg max basis) relative to the Ra 20 rotor at each of the 14 matched fan set points. Horizontal lines: geometric-mean change. Shaded bands: 95% intervals that assume independent set points (Section 3.4 gives the wider intervals that allow for correlation).

1 Objective and scope

The question is whether the surface texture of 3D-printed blades changes the power a small vertical-axis wind turbine (VAWT) can deliver, and by how much. This report covers every texture test run on the automated rig to date. It places the Jeong-lab tests of June and July in context, states what the data do and do not support, and lists the measurements that would close the remaining gaps. The accompanying data package contains every raw file used, unmodified, together with a data dictionary, checksums and the analysis code (Appendix D). Every number in this report is generated from those files.

2 Test articles

2.1 Rotor

The rotor is a three-bladed VAWT with straight blades on arms (Table 1). Its blade section is not an airfoil. Each blade is a thin cambered plate: a constant 1.79 mm wall, about 24.5 mm deep overall on a 48 mm chord, 183° of turning and square-cut edges, prismatic along the span with no twist. In cross-section each blade is closer to a half-cylinder bucket than to a wing. Because a VAWT sweeps a cylinder, the swept area is $A = 2RH$.

2.2 The texture specimens, and what is known about each

All textured rotors are printed copies of geometry v1, the first printed replica of the original rotor, which the professors chose over a remodelled v2 on 4 August. They differ in the slicer’s fuzzy-skin setting, which randomly displaces the outer wall. The controlled variable is the fuzzy-skin *thickness*, and it doubles from one rotor to the next (Table 2).

Four facts about the specimens limit how the results can be read.

Table 1. Rotor geometry (common to all test rotors; from the blade mesh).

Configuration	3 straight blades, zero twist
Radius R (axis to blade attachment)	0.1016 m
Span H	0.2451 m
Swept area $2RH$	0.0498 m ²
Chord c	48 mm
Section	cambered plate, 1.79 mm wall, square edges, 183° turning
Chord Reynolds number	3.3×10^4 to 1.2×10^5 (nominal wind, air at 20 °C)

Table 2. The rotors in this report. Thicknesses of the textured sets were confirmed by the operator. The Ra labels are not measurements (see text).

Rotor	Texture	Tested	Known / inferred
Initial reference	—	5 Jun (lab)	Probably the rotor used before the printed replicas, which were finished 13 July. Not confirmed.
No texture	none	27 Jul (lab)	Printed replica without fuzzy skin (“the new blades which have no texture”). Geometry probably v1; to be confirmed by T. Kang.
Ra 20	0.050 mm	20 Aug (rig)	Printed in the garage on a Bambu printer with a 0.4 mm nozzle and 0.08 mm layers. The run was first logged as v2_Ra20 and relabelled; the data are unchanged.
Ra 40	0.101 mm	1 Sep (rig)	Same garage print batch as Ra 20. Carried a small magnet on one blade for a reed-switch rotor-speed sensor.
Ra 80	0.202 mm	26 Aug (rig)	Printer, nozzle and layer not recorded.

- **The Ra values are labels, not measurements.** “Ra 20” and “Ra 40” are the roughness targets that an AI assistant (ChatGPT) suggested on 4 August for thicknesses of 0.050 and 0.101 mm. That same table put 0.025 mm at Ra 10. “Ra 80” was never in that table: it extends the doubling. A second AI estimate (12 August) put the 0.050 and 0.101 mm settings at about 10–15 and 20–35 μm , one-half to three-quarters of the labels. No surface has been measured. This report uses fuzzy-skin thickness as the texture variable.
- **The print details typed into the run notes are not reliable.** The notes say “0.2mm layer” and, for Ra 80, “0.2mm nozzle”. The garage printer had only 0.4 mm nozzles throughout the campaign, and the person who printed the sets reported 0.08 mm layers. The slicer project files are archived but have not yet been read.
- **The prints came from more than one source.** Nothing records which print file produced which rotor, and on 13 August two print sets were noted to “feel very different”. If the rotors differ in mass, wall thickness or edge finish as well as in texture, those differences are confounded with texture here.
- **The Ra 40 rotor carried a small magnet** on one blade for the reed switch; the others, as far as the records show, did not.

3 Facility, instrumentation and protocol

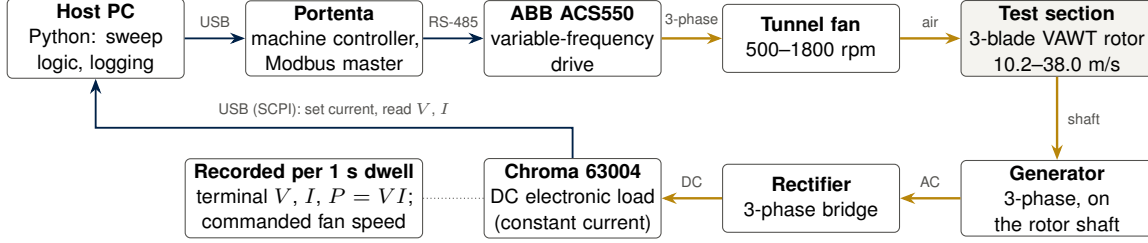


Figure 2. Measurement and control chain used for the rotor sweeps. Rotor speed was not recorded.

3.1 Rig

Fig. 2 shows the chain used for the rotor sweeps. A Python host commands the tunnel fan through a Portenta Machine Control and an ABB ACS550 drive. The generator on the rotor shaft feeds a three-phase bridge rectifier. A Chroma 63004 DC electronic load, controlled by the same host, sets the load current and measures terminal voltage and current. Wind speed comes from the tunnel calibration $v = 0.02132 N_{\text{fan}} - 0.424$ (m/s, fan speed N_{fan} in rpm, $R^2 = 0.9996$), applied to the *commanded* fan speed (Section 4).

3.2 Sweep protocol

All rig runs used one automated protocol. The software identifies it by the fingerprint 94bed28333f7, which it computes from every setting that affects what a measurement means. For each of 14 fan set points from 500 to 1800 rpm in steps of 100:

1. the fan is commanded, and the host waits for the drive to reach speed and for the terminal voltage to settle;
2. the load steps its constant-current demand upward, dwelling 1.0 s per step and recording voltage and current at the end of each dwell. The step is 20 mA at 1800 rpm and scales with v^2 at lower wind, so every wind speed gets a comparable number of steps;
3. the ramp stops at the first step where power falls below 80% of the running maximum (the last step in these runs was at 54–79% of the peak), the load is released, and the fan steps up.

A run takes about eight minutes.

3.3 Peak power: two estimators

Two estimates of P_{max} come from each load ladder.

- The *raw arg max* is the largest single dwell.
- The *parabolic fit* is the vertex of a least-squares parabola through the arg max and the two dwells either side.

The top of $P(I)$ is flat, so the arg max is biased slightly high. The Ra 20 run predates the logged fit, so we recomputed both estimators for every run from its raw dwells, using the rig’s own rule (Appendix B). The recomputed arg max values equal the logged ones exactly. The recomputed fits match the logged Ra 40 and Ra 80 fits to within 0.14%, which is the rounding of the stored values. The fit fell back to the arg max at 2 set points (Ra 20 at 700 rpm and Ra 40 at 600 rpm). The headline numbers use the raw arg max; the two bases agree (Table 4).

Table 3. All texture-campaign data.

Condition	Date	Source	Status
Ra 20	20 Aug	rig	14/14 set points clean, 336 dwells; predates the fit column and timestamps
Ra 80	26 Aug	rig	14/14 clean, 352 dwells
Ra 40	1 Sep	rig	14/14 clean, 352 dwells; also air temperature and pressure; reed-switch rotor speed not usable
No texture	27 Jul	Jeong-lab DAQ	360 Hz raw export and summary table (Section 6)
Initial reference	5 Jun	Jeong-lab DAQ	summary table and plots; raw export not in hand
Ra 10	—	—	planned, not run (1 Sep tested Ra 40 instead)

3.4 Statistics

Rotors are compared at *matched fan set points*. For each pair the change is the geometric mean of the 14 paired ratios. Two 95% intervals are given.

- The first uses the t distribution on the log-ratios and assumes the 14 set points are independent.
- They are not. Neighbouring set points share local deviations: the lag-1 autocorrelation of the log-ratios is 0.49 for Ra 80 against Ra 20. The second interval therefore uses an AR(1) effective sample size, and it is the one quoted in the Summary.

Both intervals describe scatter between wind speeds within one mounting of each rotor. Neither includes mount-to-mount or print-to-print variation. Counts such as “higher at 14 of 14” are likewise not independent replications: a single offset over a whole run would produce them. The change in the power-law exponent, Δn , is the slope of the paired log-ratio against $\ln v$.

4 Data inventory and quality

Every set point of every rig run finished cleanly. The load ramp stopped on the power roll-off each time, never on the current ceiling or a rotor stall, so every $P_{\max}$ is a true maximum rather than a lower bound. All rig runs carry protocol 94bed28333f7.

Within-mounting repeatability is good. Two single-point re-measurements of Ra 20 on the same mounting (on a different protocol, so indicative only) matched the sweep to within 0.3% at 1200 and 1800 rpm.

Fan-speed logging changed between runs; the fan speed did not. The Ra 20 file records the fan 4–21 rpm below its command, while Ra 40 and Ra 80 record it within 0–1 rpm. For Ra 20 the software logged the drive’s *output frequency* multiplied by a fixed 29.5 rpm/Hz, and all 14 values fall on that 0.1 Hz grid. From 25 August it read the drive’s own speed estimate. The fan motor current at matched command agrees across the three runs to within 0.1 A, which a fan running up to 21 rpm slower would not do. All comparisons are therefore made at matched *commanded* speed. As a bound only: if the Ra 20 readback *had* been a real air-speed deficit, correcting for it would reduce the gains to +4.3% and +9.4%.

Ambient conditions were recorded only for Ra 40 (24.6 °C, 101.04 kPa, sensor offset uncalibrated). On this rig $P_{\max}$ rises faster than linearly with dynamic pressure, so a density difference of a few percent between days could shift $P_{\max}$ by up to about twice that. This is small next to the Ra 80 gain but comparable to the Ra 80-vs-Ra 40 step.

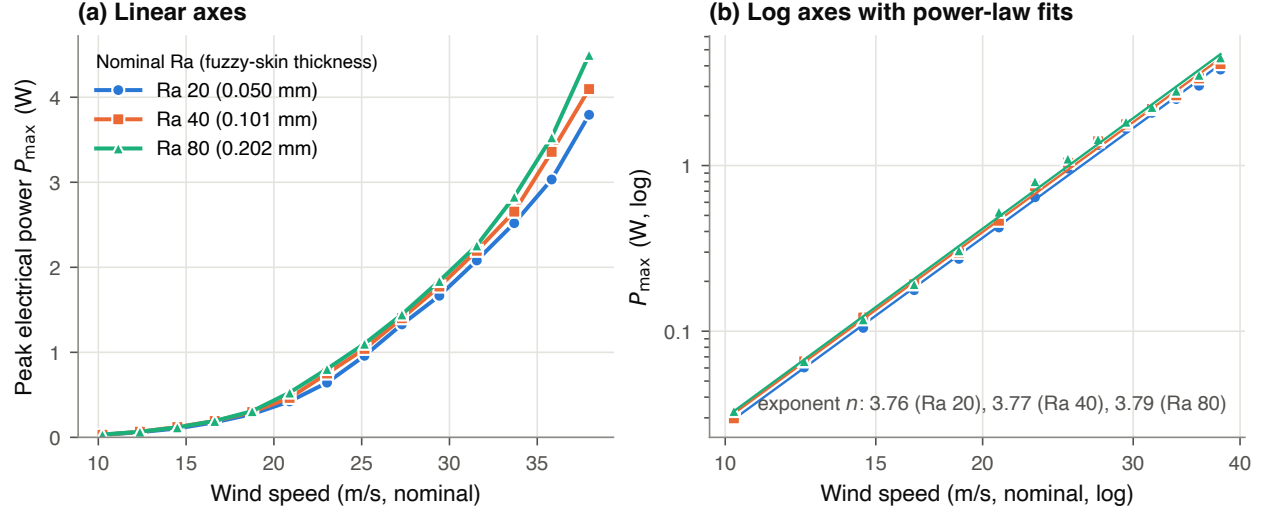


Figure 3. Peak electrical power (raw arg max) against nominal wind speed. (a) Linear axes. (b) Log axes with a power-law fit per rotor.

Table 4. Paired comparisons at the 14 matched fan set points. CI: 95% interval assuming independent set points. CI_{AR} : allowing for correlation between neighbouring set points. Δn : change in the power-law exponent, $\pm$ 95% half-width. Changes in percent.

Comparison	Basis	Change	CI	CI_{AR}	Higher	Range	Δn
Ra 40 vs Ra 20	raw arg max	+8.4	[+6.1, +10.8]	[+5.7, +11.2]	14/14	+0.3 to +16.7	+0.003 $\pm$ 0.058
Ra 80 vs Ra 20	raw arg max	+13.7	[+10.7, +16.9]	[+6.9, +21.0]	14/14	+8.5 to +25.0	+0.024 $\pm$ 0.071
Ra 80 vs Ra 40	raw arg max	+4.9	[+2.4, +7.4]	[+1.5, +8.5]	13/14	-2.5 to +13.4	+0.021 $\pm$ 0.062
Ra 40 vs Ra 20	parabolic fit	+8.7	[+6.1, +11.4]	[+6.1, +11.4]	13/14	-1.3 to +18.7	+0.024 $\pm$ 0.063
Ra 80 vs Ra 20	parabolic fit	+13.7	[+9.9, +17.5]	[+6.8, +21.0]	14/14	+6.6 to +28.4	+0.048 $\pm$ 0.083
Ra 80 vs Ra 40	parabolic fit	+4.5	[+2.1, +7.1]	[+1.3, +7.9]	12/14	-2.4 to +15.8	+0.024 $\pm$ 0.062

5 Results

5.1 Power curves

Fig. 3 shows $P_{\max}$ against wind speed. Peak power rises from about 0.03 W at 10.2 m/s to 3.79, 4.10 and 4.50 W at 38.0 m/s for Ra 20, Ra 40 and Ra 80. Each curve follows $P_{\max} \propto v^n$ with $n = 3.76$, 3.77 and 3.79 (Table 5). Appendix A tabulates every set point.

5.2 Paired comparison

Table 4 and Fig. 1 give the paired comparisons. Individual set points scatter by about 4%: the standard deviation of the log-ratios is 3.8% for Ra 40 and 4.7% for Ra 80. A single set point can therefore show anything from +8.5% to +25.0% for Ra 80.

The largest deviations are at fan 1000–1100 (21 m/s to 23 m/s). At 1100 rpm both textured rotors are high (+16.7% and +25.0% against Ra 20), which points to a low Ra 20 reading there. At 1000 rpm only Ra 80 is high: it is +24.6% over Ra 20 and +13.4% over Ra 40, so that point is specific to Ra 80.

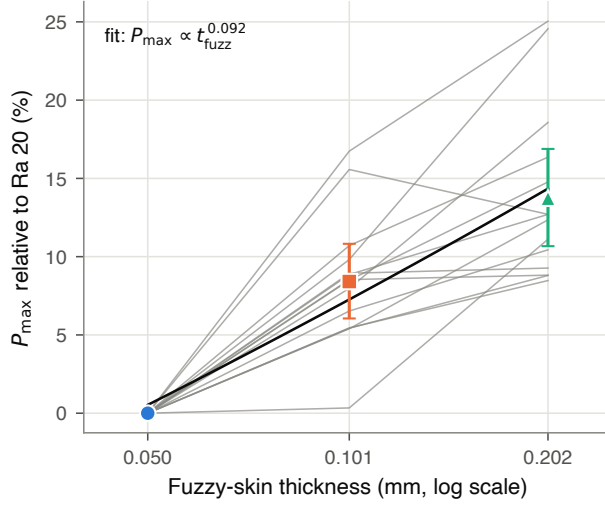


Figure 4. Peak power relative to Ra 20 against fuzzy-skin thickness. Grey lines: individual set points. Markers: geometric mean with 95% interval (independent set points). Black line: the single-slope fit.

5.3 Trend with texture

Power increases with fuzzy-skin thickness at 13 of 14 set points (Fig. 4). The exception is 700 rpm, where Ra 40 and Ra 80 are within 2.5%. A single slope in log-log coordinates gives $P_{\max} \propto t_{\text{fuzz}}^{0.092}$, about +6.6% per doubling of thickness. No uncertainty can yet be attached to that slope. It rests on three rotors mounted once each, and a 3% offset on the single Ra 20 mounting would move it by about a quarter.

5.4 Where the extra power comes from

At a fixed wind speed the rotor, generator and rectifier together behave as a Thévenin source. Across each load ladder the terminal voltage falls linearly with current, $V = V_{\text{oc}} - IR_{\text{int}}$, with $r^2 \geq 0.975$ at every set point of every run.

- V_{oc} is the zero-current intercept. It tracks the generator EMF, and hence rotor speed at light load, to within a rectifier offset of about 0.45 V.
- R_{int} combines the winding, rectifier and wiring with the rotor's slowing under load.
- At the power maximum $P_{\max} \approx V_{\text{oc}}^2/4R_{\text{int}}$, so a change in power can be split between the two.

Fig. 5 and Table 8 show the result.

- **Open-circuit voltage.** The textured rotors' V_{oc} is higher at every one of the 14 set points: by +4.1% ([+3.3, +4.9]) for Ra 40 and +6.4% ([+4.9, +7.9]) for Ra 80.
- **Source resistance.** Their R_{int} is indistinguishable from Ra 20's: -0.7% ([-2.7, +1.4]) and -0.4% ([-2.2, +1.5]). Within its uncertainty, the V_{oc} change accounts for the whole power gain.
- **A direct check.** The voltage measured at the first, lightest load step, with no fitting, is higher by +3.9% and +6.3%, at all 14 set points for both rotors.

At light load the textured rotors turn faster in the same wind, and the extra power follows from that. The effect is on the rotor side of the generator, not in the electrical path. The pattern differs between the two rotors. The Ra 40 advantage is flat across the range (+4.1% below 17 m/s, +3.7% above 31 m/s). The Ra 80 advantage grows with wind speed, from +3.5% to +7.8%: against Ra 40 its slope in $\ln v$ is +0.042 ($p = 0.003$). Faster free-running does not by itself identify the mechanism. More aerodynamic torque and less parasitic torque would both produce it (Section 7).

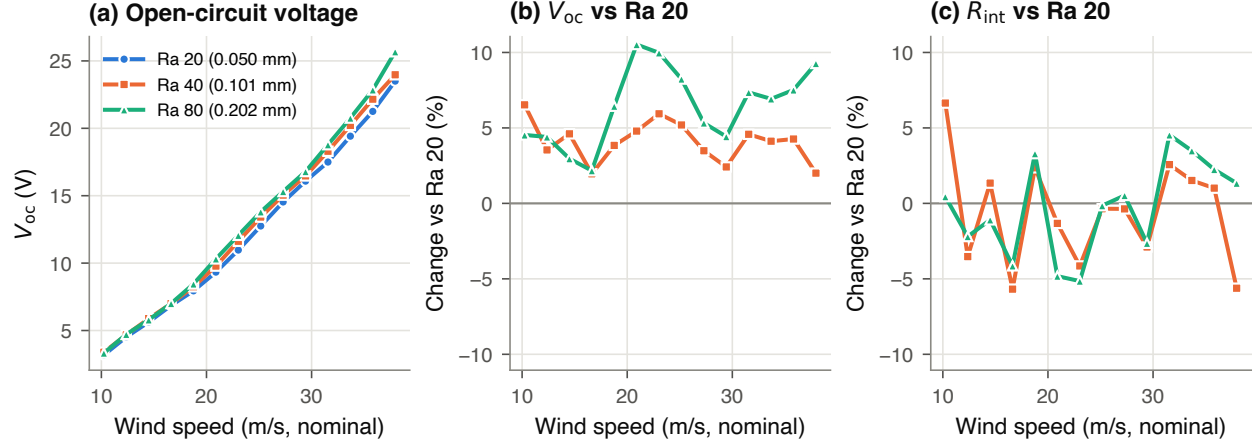


Figure 5. Thévenin decomposition of every load ladder. (a) Open-circuit voltage. (b) Open-circuit voltage and (c) apparent source resistance, each relative to Ra 20.

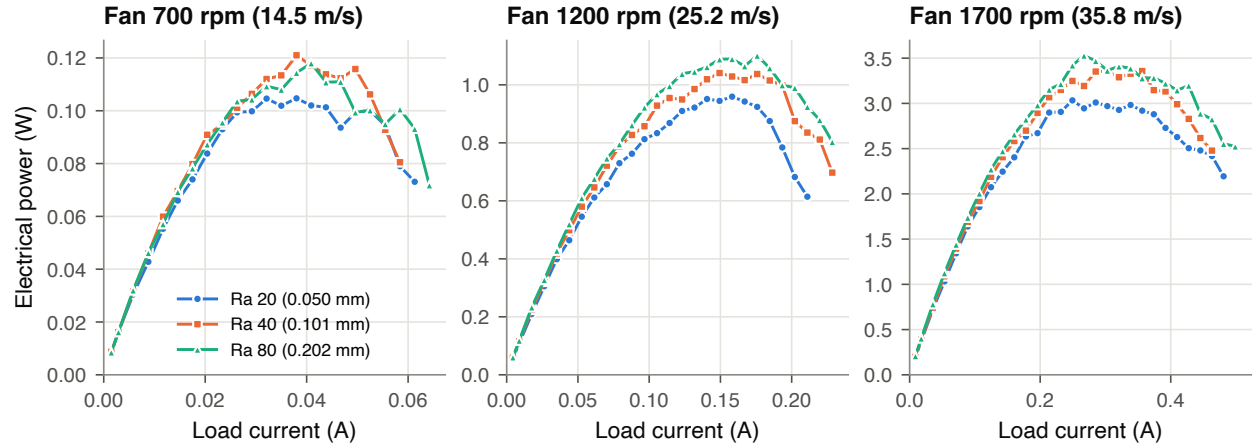


Figure 6. Measured power–current ladders at three wind speeds. The textured rotors are higher across the whole ladder, not only near the peak.

5.5 Scaling with wind speed

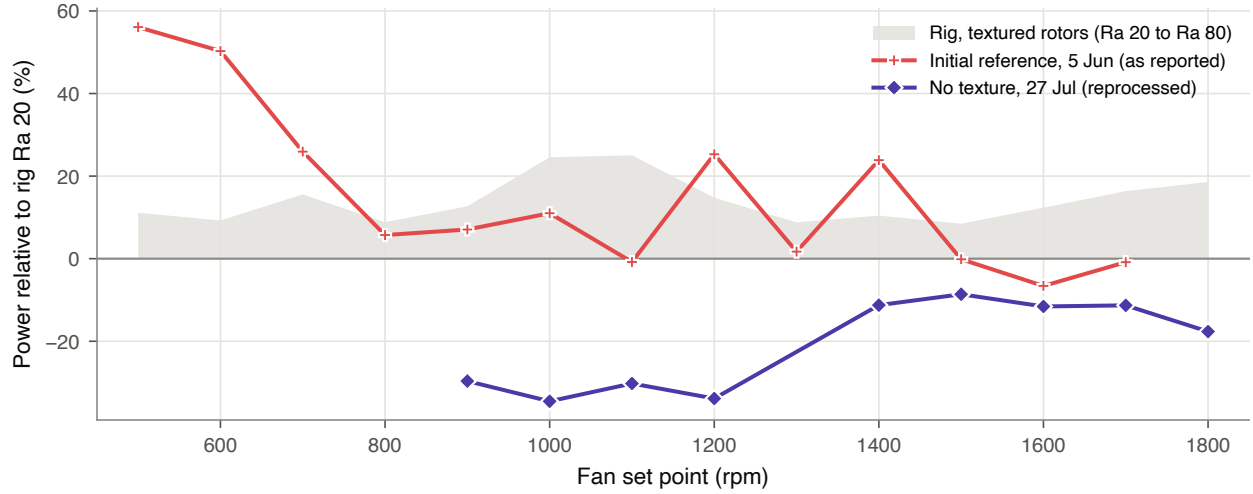
All rotors follow $P_{\max} \propto v^n$ with n between 3.76 and 3.79 (Table 5). The paired exponent changes are zero within their intervals: $\Delta n = +0.003 \pm 0.058$ for Ra 40 and $+0.024 \pm 0.071$ for Ra 80.

That rules out one alternative explanation. Suppose the rotors differed in cut-in speed, for example through constant friction. A cut-in model fitted to Ra 20, $P_{\max} = a(v - v_c)^m$ with $v_c = 0.8$ m/s and $m = 3.60$, needs a shift of 0.72 m/s to reproduce the whole Ra 80 gain. That shift would change n by -0.14 , outside the measured interval. For Ra 40 the shift would be 0.45 m/s and the exponent change -0.09 , excluded only narrowly. A partial cut-in contribution of a few percent cannot be excluded.

$n \approx 3.8$ exceeds 3 because free-running speed rises faster than wind speed ($V_{oc} \propto v^{1.5}$) and the rotor slows less under load at higher wind ($R_{int} \propto v^{-0.79}$). Both are properties of the rotor, including its friction. That $n \approx 2(1.50) + 0.79$ follows algebraically from the Thévenin form. It is a consistency check, not evidence about which component sets the exponent.

Table 5. Power-law fits $P_{\max} = a v^n$ against nominal wind speed (a in μW per $(\text{m/s})^n$).

Rotor	Basis	n	95% CI	a	R^2 (log)	R^2 (linear)
Ra 20	raw arg max	3.763	[3.669, 3.856]	4.662	0.9984	0.9899
Ra 20	parabolic fit	3.754	[3.660, 3.848]	4.732	0.9984	0.9900
Ra 40	raw arg max	3.766	[3.667, 3.864]	5.008	0.9983	0.9902
Ra 40	parabolic fit	3.778	[3.676, 3.880]	4.772	0.9982	0.9903
Ra 80	raw arg max	3.787	[3.684, 3.890]	4.920	0.9981	0.9934
Ra 80	parabolic fit	3.802	[3.693, 3.912]	4.631	0.9979	0.9924

**Figure 7.** The Jeong-lab tests relative to the rig’s Ra 20 at the same fan set point. Grey band: the range spanned by the rig’s three textured rotors. The 27 July values are reprocessed from the raw export (Appendix C); 500–800 rpm and the 1300 rpm window are omitted (no load current, or a window spanning a speed step). 5 June: as tabulated by the lab (probably the original rotor).

6 Relation to the Jeong-lab tests

Two earlier datasets bear on the question: the *initial reference* of 5 June and the *no-texture* test of 27 July. Both predate the automated rig, whose first run was on 20 August, and both were recorded on the Jeong-lab DAQ. The two setups use slightly different fan-to-wind calibrations (9.79 m/s against 10.24 m/s at 500 rpm), so this section compares at matched fan set point. Fig. 7 shows both datasets relative to the rig’s Ra 20.

6.1 Comparison with the initial test

Prof. Jeong asked on 27 July that new results be compared with the initial test. At 800–1700 rpm the 5 June values lie between -6.6% and $+25.3\%$ of the rig’s Ra 20 (geometric mean $+6.2\%$). At 500–700 rpm, where powers are below 0.15 W, they are $+25.9$ to $+56.1\%$ higher. Over the whole range they sit about level with the rig’s Ra 80 (geometric mean $+0.4\%$). The printed rotors therefore perform about like the initial reference. The rotor, probably the original one, and the processing differ, so this is a consistency check, not a texture comparison. Putting the June data on the same basis as the rest would need the June raw export, which we would be glad to process together with Taegu.

6.2 The 27 July no-texture test

We ran this test with the lab’s DAQ recording at 360 Hz. Because Taegu shared the complete raw export, the lab’s `Summary_Table.csv` can be rebuilt from it exactly: all 64 values reproduce to floating-point precision (Appendix C). The rebuild shows two things that matter for comparison with the rig.

- **The table’s columns are maxima of raw 360 Hz samples.** The column names suggest the aim was the maximum power at each setting. Over raw samples, a maximum also collects channel noise, which here raises P by about $0.11 \text{ A} \times V$. The rig reads one settled value per 1 s dwell.
- **At 500–700 rpm the load was off.** The tabulated power there reflects channel noise rather than delivered power. Those windows do give a clean free-running voltage: 3.11, 3.84 and 5.39 V. The rig’s Ra 20 reads 2.85, 4.47 and 5.62 V at its first load step. This comparison needs no current calibration, and it shows the two setups agree on rotor speed to within about 15%.

We then reprocessed the same windows on the rig’s basis: current zeroed at load-off, and a 1 s mean before taking the maximum (the settled mean would be 2–11% lower still). On that basis the no-texture values are below the rig’s Ra 20 at all 9 usable set points from 900 to 1800 rpm, by 9–18% at 1400–1800 rpm. This cannot be read as a texture effect:

- the two tests differ in print, mounting, day and electrical chain;
- the current scale (2 A/V , i.e. a 500 mV/A sensor) cannot be checked from the file, and a comparison of the lab’s and rig’s voltage–current lines suggests it may under-read, by enough to remove or reverse the gap.

What the July test can support is narrower: it cannot say whether the un-textured rotor is above or below the textured ones at the 8–14% level.

6.3 A joint session would settle this

On 1 October the lab’s DAQ can record in parallel with the Chroma load during a rig sweep. That would do three things:

- calibrate the lab’s current channel directly against the Chroma;
- show both statistics on the same data;
- add the lab’s once-per-revolution pulse, which tracked the rotor cleanly at every setting up to 1900 rpm. It would give the rig the rotor speed it lacks for λ and $C_p(\lambda)$.

Questions for Taegu:

1. What is the current sensor’s part number, range and zero?
2. What is the input range of the voltage channel?
3. How is the pulse generated?

The recipe in Appendix C is offered for checking against the lab’s own script. Please correct anything we have inferred wrongly.

7 Discussion

What is established. On one mounting of each rotor, peak electrical power increases with fuzzy-skin thickness over 0.050 mm to 0.202 mm, and the increase is carried by the rotor’s free-running speed rather than the electrical path.

The operating regime. The Jeong-lab pulse gives a loaded tip-speed ratio of $\lambda \approx 0.12\text{--}0.16$. Mapping the rig's V_{oc} through the generator constant from the lab's unloaded windows gives a free-running $\lambda \approx 0.11\text{--}0.22$. Both are estimates, since the rig did not record rotor speed. Even unloaded, then, the rotor turns far slower than a lift-driven rotor ($\lambda \approx 3\text{--}5$): with its 183° bucket-like blades it behaves as a drag-driven machine. This, more than generator loading, explains why the electrical power coefficient at peak, $P_{\max}/(\frac{1}{2}\rho Av^3)$ with standard air density, is only $0.09\text{--}0.27\%$.

Mechanism. In a drag rotor the net torque is the difference between the drag on the concave (advancing) and convex (returning) faces. On the concave face, separation is fixed by the sharp edges. On the convex face it is not, and surface roughness can move it. That is the golf-ball effect, and it would reduce the drag on the returning blade. The Ra80 voltage advantage growing with wind speed (Section 5.4) is what such a Reynolds-dependent effect would produce. It is a candidate, not a demonstrated mechanism. Fuzzy skin also changes the printed shape slightly, which is a second, geometric path. The Sandia study circulated in August [1] concerns airfoils at utility-scale Reynolds numbers, where roughness is detrimental, so it is not the right reference for this rotor.

Alternative explanations, and how far the data exclude them.

- *A one-off change between the first run and the others.* Test order was Ra 20 (20 Aug), Ra 80 (26 Aug), Ra 40 (1 Sep). That excludes a steady drift with date, but not a step. Ra 20 is the only run taken before a round of software and firmware changes and a drive-parameter restore (22–26 Aug), so a step would lift both textured rotors above it. The comparison free of this confound is Ra 80 against Ra 40: $+4.9\%$ ($[+1.5, +8.5]$), with the Ra80 voltage advantage growing with wind. The remount of Ra 20 on current software tests the step directly.
- *Friction and mounting.* A constant-friction difference large enough to explain the gains would change the exponent (Section 5.5), and is excluded for Ra 80, narrowly for Ra 40. Losses that rise with rotor speed (viscous bearing drag, imbalance) change the exponent less, and the present intervals cannot exclude them. Remounting, and a spin-down test, would.
- *Print differences other than texture.* Mass, wall thickness, edge finish and the Ra 40 magnet (Section 2). Weighing the blades and reading the slicer files would bound these.
- *Wind speed on the day.* There is no independent wind measurement. A stronger wind would raise V_{oc} and also lower R_{int} : by about -3.2% for the Ra 80 voltage change, against an observed -0.4% ($[-2.2, +1.5]$). That argues against a wind offset explaining *all* of the Ra 80 gain, but allows one explaining part of it. For Ra 40 the test does not discriminate.
- *Fan-speed logging and air density.* Discussed in Section 4.

Electrical power is not aerodynamic efficiency. The Thévenin analysis shows the textured rotors spinning faster unloaded, but lower parasitic torque would produce exactly that, and the rotors are compared at different tip-speed ratios. It is not a substitute for a $C_p(\lambda)$ comparison, which needs a trustworthy rotor-speed channel.

8 Limitations

9 Recommended next steps

In order of value per tunnel-hour:

1. **Remount each rotor, in randomised order.** One remount of one rotor can detect a large mounting effect but cannot show that it is small, because it gives a single difference. If a remount moves $P_{\max}$ by more than about 3% , the Ra 40-vs-Ra 20 and Ra 80-vs-Ra 40 differences are not

Table 6. Limitations, their likely effect, and what would resolve them.

Limitation	Effect on the conclusions	Resolved by
One mounting per rotor	Intervals omit mount-to-mount variation	More mountings, randomised order
Ra 20 predates software changes	A one-off step would inflate both gains	Remount Ra 20 on current software
Texture not measured	Ra values are labels; thickness is the reliable variable	Profilometer Ra per blade set
Print details unverified	Nozzle, layer, mass, edges or the Ra 40 magnet may co-vary with texture	Read slicer files; weigh blades
No rotor speed on the rig	Electrical power only; no measured λ or $C_p(\lambda)$	Lab DAQ pulse in parallel; Hall sensor
No independent wind measurement	A partial day-to-day wind offset cannot be excluded	Log the anemometer on every run
Ambient not recorded (Ra 20, Ra 80)	Density differences could shift $P_{\max}$ a few percent	Calibrated temperature and pressure every run
No un-textured rotor on the rig	Cannot say whether texture helps relative to none	Run the no-texture rotor on the rig

established. Establishing the ordering needs a mount-to-mount standard deviation below about 1.5–2%, measured from at least two remounts of each rotor.

2. **Run the no-texture rotor on the rig**, and Ra 10, to anchor the texture axis.
3. **Record the lab DAQ in parallel** during a rig sweep (Section 6). This calibrates the current channel and adds rotor speed.
4. **Characterise the specimens**: read the slicer files, weigh each blade set, and measure Ra.
5. **Log ambient air and the anemometer** on every run.

9.1 Toward the fault-testing plan

Prof. Jeong asked in July for a fault-testing plan, agreed before the next round of tests. It has not been drafted. The campaign so far suggests a starting point for that discussion.

- **What the rig offers.** An automated peak-power sweep takes about 8 minutes. The Thévenin decomposition separates rotor-side changes, which show in V_{oc} , from electrical ones, which show in R_{int} . Changes of about 4% in V_{oc} were resolved at every set point. The tunnel node's accelerometer can add vibration spectra for imbalance.
- **Candidate faults to discuss.**
 - A tip mass on one blade (imbalance).
 - A chipped or shortened blade, or a missing blade.
 - Added bearing drag.
 - An open or high-resistance generator phase.
- **Proposed next step.** A short meeting to agree the fault list, its severity levels and the acceptance criteria before any fault blades are printed.

References

- [1] B. Wilcox, E. White and D. Maniaci, “Roughness sensitivity comparisons of wind turbine blade sections,” Sandia National Laboratories, 2017. OSTI 1404826.

A Per-set-point data

Table 7. Peak electrical power (W) by rotor and estimator. Wind speed is nominal, from the commanded fan speed.

Fan rpm	v m/s	Ra 20		Ra 40		Ra 80	
		raw	fit	raw	fit	raw	fit
500	10.2	0.0297	0.0296	0.0298	0.0293	0.0330	0.0316
600	12.4	0.0604	0.0591	0.0658	0.0658	0.0660	0.0657
700	14.5	0.1047	0.1047	0.1210	0.1186	0.1180	0.1158
800	16.6	0.1769	0.1761	0.1920	0.1851	0.1925	0.1877
900	18.8	0.2732	0.2743	0.2974	0.2965	0.3079	0.3023
1000	20.9	0.4227	0.4088	0.4642	0.4532	0.5265	0.5248
1100	23.0	0.6433	0.6320	0.7510	0.7500	0.8044	0.7990
1200	25.2	0.9592	0.9542	1.0407	1.0375	1.1009	1.0917
1300	27.3	1.3286	1.3208	1.4007	1.3820	1.4455	1.4395
1400	29.4	1.6653	1.6225	1.7740	1.7710	1.8391	1.8170
1500	31.6	2.0806	2.0454	2.1940	2.1953	2.2569	2.2446
1600	33.7	2.5188	2.4795	2.6547	2.6427	2.8299	2.7846
1700	35.8	3.0329	2.9966	3.3576	3.3248	3.5290	3.5181
1800	38.0	3.7935	3.7217	4.0961	4.1018	4.4981	4.4317

B Methods detail

Peak power. For each set point, the dwells in which the load tracked its demand with current above zero form the ladder (I_k, P_k) . The raw estimate is $\max_k P_k$. The fit proceeds as follows.

1. Take the $\arg \max k^*$ and the dwells $k^* - 2 \dots k^* + 2$ (fewer at the ends).
2. Fit $P = a_2 I^2 + a_1 I + a_0$ to them by least squares.
3. Report $P(\hat{I})$ at $\hat{I} = -a_1/2a_2$, provided $a_2 < 0$ and $\hat{I}$ lies inside the window; otherwise report the $\arg \max$.

This is a direct port of the rig software’s rule.

Paired change. With $d_i = \ln(P_{c,i}/P_{b,i})$ over the 14 set points, the change is $\exp(\bar{d}) - 1$. The independent-set-point interval is $\exp(\bar{d} \pm t_{0.975,13} s_d/\sqrt{14}) - 1$. The AR(1) interval replaces 14 with $n_{\text{eff}} = 14(1 - r_1)/(1 + r_1)$, where r_1 is the lag-1 autocorrelation of d_i ordered by set point.

Exponent change. Δn is the least-squares slope of d_i against $\ln v_i$, with a $t_{0.975,12}$ interval.

Texture trend. $\ln P_{r,s} = \alpha_s + \beta \ln t_{\text{fuzz},r}$, with a free intercept per set point and one slope. Rotors whose identity is unconfirmed are excluded.

Thévenin fit. Per set point, $V = V_{\text{oc}} - IR_{\text{int}}$ by least squares over all tracking dwells with $V, I > 0$, as in the rig software’s generator model.

Table 8. Thévenin fit per set point: open-circuit voltage V_{oc} (V) and apparent source resistance R_{int} (Ω).

Fan	v	Ra 20		Ra 40		Ra 80	
		V_{oc}	R_{int}	V_{oc}	R_{int}	V_{oc}	R_{int}
rpm	m/s						
500	10.2	3.18	88.4	3.39	94.2	3.33	88.8
600	12.4	4.52	87.1	4.68	84.0	4.72	85.2
700	14.5	5.62	73.6	5.88	74.6	5.79	72.8
800	16.6	6.86	67.5	6.99	63.7	7.01	64.7
900	18.8	7.94	58.3	8.25	59.7	8.45	60.3
1000	20.9	9.33	54.9	9.78	54.2	10.31	52.2
1100	23.0	10.97	49.3	11.62	47.3	12.06	46.8
1200	25.2	12.75	44.4	13.42	44.3	13.80	44.3
1300	27.3	14.54	41.4	15.04	41.3	15.31	41.6
1400	29.4	16.08	39.6	16.47	38.4	16.79	38.5
1500	31.6	17.50	36.8	18.30	37.7	18.78	38.4
1600	33.7	19.42	36.6	20.22	37.2	20.76	37.9
1700	35.8	21.26	36.2	22.16	36.6	22.85	37.1
1800	38.0	23.50	36.5	23.97	34.4	25.68	36.9

Cut-in model. $P_{\max} = a(v - v_c)^m$ is fitted to Ra 20 by scanning v_c . The shift in v_c that reproduces each observed gain, and the exponent change it implies, are computed from that fit.

C Reprocessing of the 27 July export

The file `0727windturbine.csv` has 127 208 rows at 360 Hz. Its columns are **Relative Time**, a date, a time stamp, five channels headed **Volt**, and an empty event column. The channel identification below was inferred from the data and should be confirmed against the lab’s wiring.

- **ch1:** the DC bus voltage through a 4:1 divider. It jumps to about 20 V when the load is switched off, then decays with an 18 s time constant, like a capacitor.
- **ch2:** a Hall-effect current sensor with a mid-rail zero.
- **ch3:** a once-per-revolution rotor pulse.
- **ch4 and ch5:** the same pulse train at lower gain.

Rebuilding `Summary_Table.csv`.

1. Let $L = \lfloor 127208/17 \rfloor = 7482$. Window $k = 0, \dots, 15$ covers 0-based samples $kL + 1870$ to $kL + 5611$ inclusive: 3742 samples, 10.39 s, the middle half of each of the first 16 of 17 equal slices.
2. $V = 4\text{ch1}$ and $I = 2(\text{ch2} - 2.5)$.
3. V_{dc_max} , I_{dc_max} and P_{dc_max} are the separate maxima of V , I and $V \cdot I$ over the window.
4. **Measured_RPM** is 60 times the number of upward jumps in ch3 larger than about 0.1 V, divided by the window duration.
5. **Wind_Speed_ms** is a lookup by fan setting.

All 64 tabulated values reproduce to floating-point precision.

Observations from the rebuild.

- **Current noise.** The current channel carries about 31 mA of white noise, the same with the load on or off. At every window’s P_{dc_max} sample the current sits 0.095–0.144 A above its 1 s

mean, so the noise adds roughly $0.11 \text{ A} \times V$ to each tabulated power. Applied to a stretch after the load was switched off, with the bus at 11–20 V, the same recipe returns 1.50 W.

- **Load schedule.** The load was held in constant current and raised by hand in a few steps per setting, and it was off for the first three settings.
- **Window alignment.** The processing assumes equal time at each setting. The settings were changed by hand (the drive automation came later), with dwells of about 12–38 s, so some windows straddle a change: the window labelled 1300 rpm contains the step to 1400 rpm.
- **Pulse count.** A few rising edges that span two samples are counted twice (five in all), and at 2000 rpm the pulse misses about one revolution in eight: true rotor speed about 667 rpm against the 595 tabulated. At 1900 rpm and below the pulse tracks cleanly.

Reprocessing used in Section 6. The same windows are used. The current zero is the median of ch2 over 331.0–341.4 s, after the load is switched off while the rotor is still turning (2.4963 V). Power $V \cdot I$ is averaged over a centred 1 s (360-sample) window, and the maximum of that average is taken within each lab window. Fig. 8 and Table 9 compare it with the table.

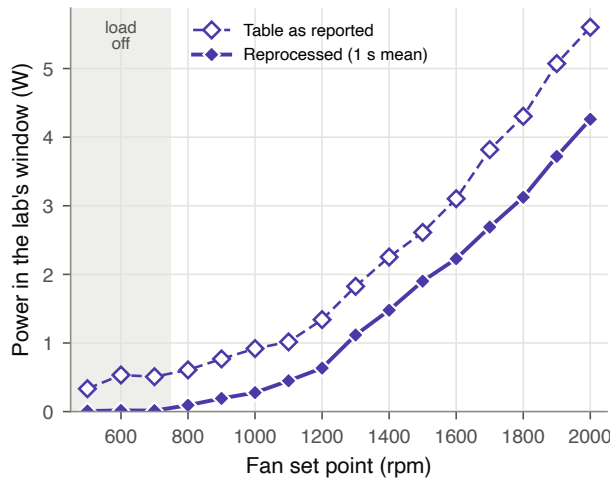


Figure 8. 27 July: the lab’s table and the same windows reprocessed with a 1 s mean and the load-off current zero. Grey: settings with the load off.

D Data package and reproducibility

The accompanying archive contains:

- `1_rig_sweeps/`: every rig file, byte for byte;
- `2_jeong_lab/`: the lab’s tables, raw export and plots, as e-mailed;
- `3_derived/`: the analysis-ready tables behind this report;
- `4_reference/`: the blade geometry;
- `5_figures/`: the figures;
- `6_code/`: the analysis code.

`README.md` explains the layout, `DATA_DICTIONARY.md` defines every column, and `MANIFEST.csv` gives each file’s SHA-256 checksum. Running `python3 6_code/build_report.py` regenerates every number, table and figure in this report from the raw files.

Table 9. 27 July no-texture test by fan set point. The columns are the lab’s **Pdc_max**; the same window reprocessed; the mean load current; the rig’s Ra 20 $P_{\max}$; and the ratio of the reprocessed value to it.

[†]Window contains the step to 1400 rpm.

Fan rpm	Lab Pdc_max (W)	Reprocessed (W)	Mean current (mA)	Rig Ra 20 (W)	Reprocessed / Ra 20
500	0.333	0.008	−1	0.030	—
600	0.533	0.017	0	0.060	—
700	0.508	0.014	0	0.105	—
800	0.607	0.093	15	0.177	—
900	0.768	0.192	28	0.273	0.70
1000	0.918	0.277	40	0.423	0.65
1100	1.016	0.449	69	0.643	0.70
1200	1.340	0.634	101	0.959	0.66
1300 [†]	1.825	1.117	127	1.329	0.84
1400	2.253	1.478	178	1.665	0.89
1500	2.611	1.901	223	2.081	0.91
1600	3.103	2.227	231	2.519	0.88
1700	3.818	2.690	271	3.033	0.89
1800	4.304	3.124	302	3.793	0.82
1900	5.073	3.720	340	—	—
2000	5.602	4.261	348	—	—